

Learning with Less: Label-Efficient Land Cover Classification at Very High Resolution

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Deep Learning for Accurate VHSR Land Cover Mapping

- DL models incorporate spatial + spectral features to produce detailed maps in an end-to-end manner (Yuan et al., 2021).
- VHSR (<5 m) land use/land cover products reveal fine-scale features critical for many applications (Li et al., 2022).
- Modern CNN/ViT models are more than capable of crossing the “resolution barrier” that plagued earlier methods (Woodcock and Strahler, 1987; Sherrah, 2016).

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Practical Barriers to VHSR Land Cover Mapping

- DL models are powerful but data-hungry (Zhu et al., 2017).
- Most research into VHSR land cover mapping use benchmark datasets with abundant labels over small areas (ISPRS, 2018b,a)
- Creating large, labeled datasets for new areas is a major bottleneck in terms of human capital (Hu et al., 2015).

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A Two-Part Approach to Label-Efficient VHSR Land Cover Mapping

How can we train accurate models with limited labeled data?

- **Part 1:** Sample smarter. Use existing, coarse land cover data to guide selection of a small but informative training set based on approximate composition of candidate patches.
- **Part 2:** Pre-train better. Use self-supervised learning on large amounts VHSR imagery to learn robust, domain-specific features without labels.

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Part 1: Smarter Sampling with Semantic Stratification

- **Idea:** Don't just sample randomly. Select patches that are representative of the diverse land cover compositions in the study area.
- **Method:** Use existing, coarse land cover data (like NLCD) to guide our sampling.

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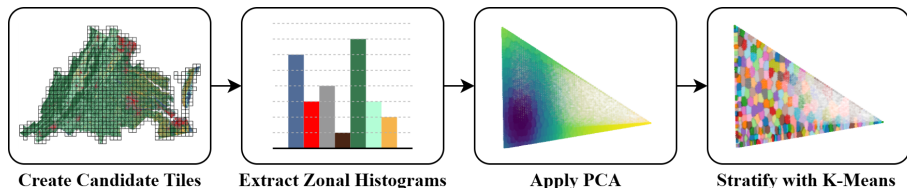


Figure 1: We use NLCD (United States Geological Survey, 2025) data to select appropriate training patches from an AOI given a grid of candidate patches.

Semantic Stratification: The Result

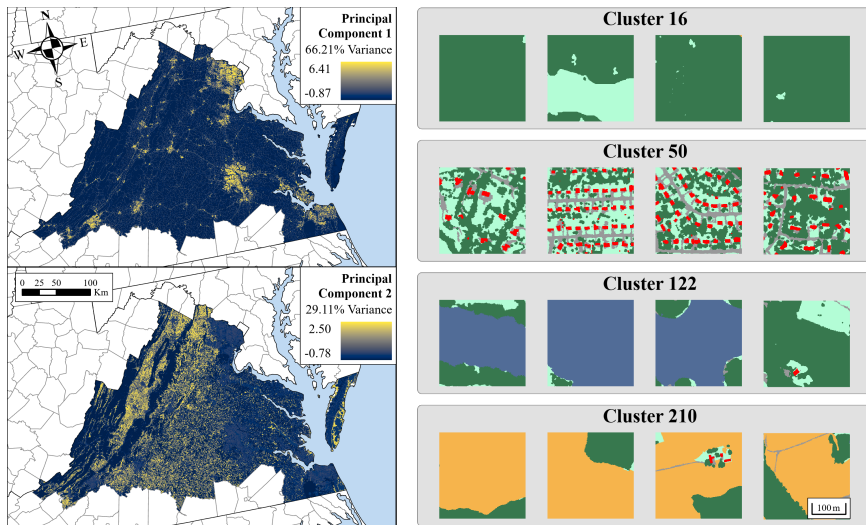


Figure 2: Patches within the same cluster tend share land cover characteristics.

Validating Our Sampling Method

- **Goal:** Test if semantic stratification is better than other methods.
- **Data:** Chesapeake Bay VHRS imagery and labels.
- **Experiment:**
 - ▶ Compare 3 sampling methods: Random, Spatial, and Semantic (ours).
 - ▶ Train DeepLabv3+ models with different training set sizes ($N = 250, 500, 750$).
 - ▶ Use rigorous statistical tests (ANOVA, Tukey's HSD) to confirm results.

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Semantic Stratification Performance

Semantic stratification leads to better models.

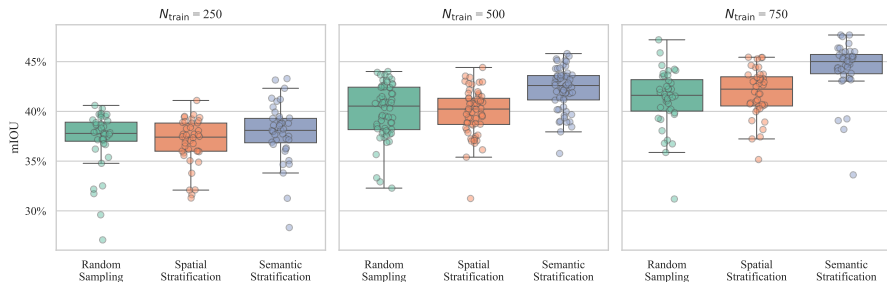


Figure 3: Our proposed method shows statistically significant performance gains over random sampling and spatially stratified sampling at $N_{\text{train}} = 500$ and $N_{\text{train}} = 750$.

Part 2: Better Pre-training with Self-Supervision

- **Idea:** Leverage vast amounts of *unlabeled* imagery to learn meaningful features before fine-tuning on our small labeled dataset.
- **Method:** Use Bootstrap Your Own Latent (BYOL), a self-supervised learning technique (Grill et al., 2020).

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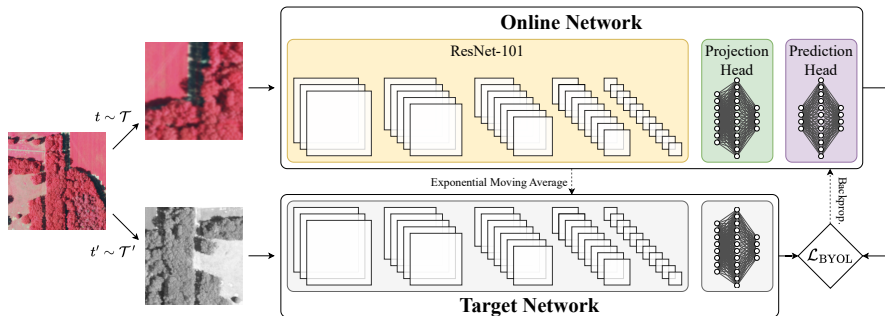


Figure 4: BYOL trains a network to predict a different augmented view of the same image, learning robust, transformation-invariant features.

Pre-Training Procedure

- Pre-train a ResNet-101 backbone with BYOL on a large set of unlabeled NAIP imagery.
- Evaluate learned features using linear probing on a small labeled set.
- Fine-tune full segmentation models (e.g., U-Net, DeepLabv3+) on labeled sets of varying sizes ($N = 250, 500, 750$).
- Compare performance against standard ImageNet pre-training.

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Assessing Pre-Trained Feature Quality

Embeddings learned via BYOL are better suited for classification under linear probing.

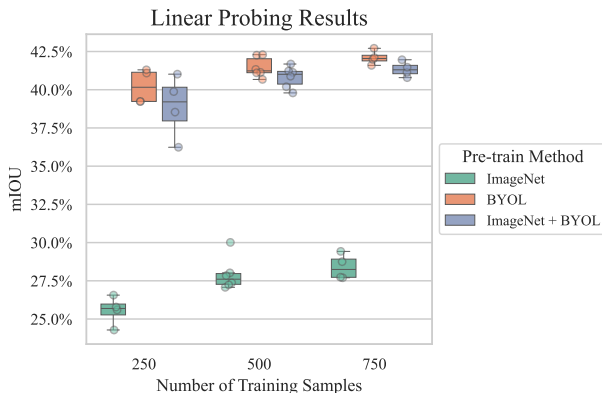


Figure 5: Linear probing shows features from BYOL models vastly outperform standard ImageNet features, closing the domain gap.

Impact on Fine-Tuning Performance

This translates to better performance on the final segmentation task.

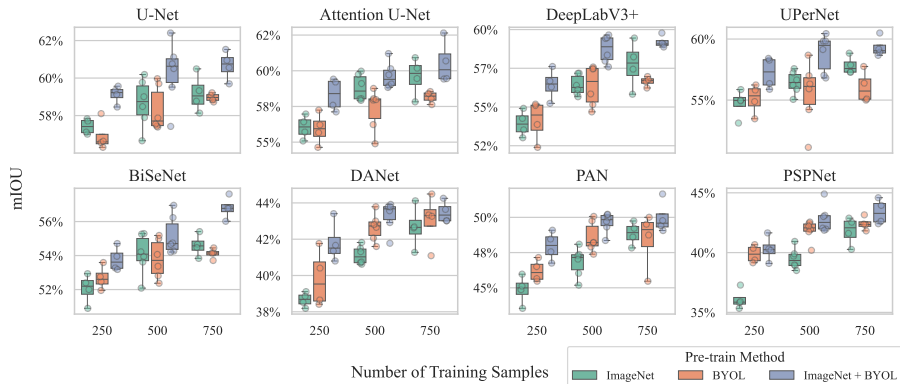


Figure 6: Fine-tuning various segmentation models shows that BYOL pre-training consistently improves mIoU, especially with few labeled samples.

Qualitative Comparison

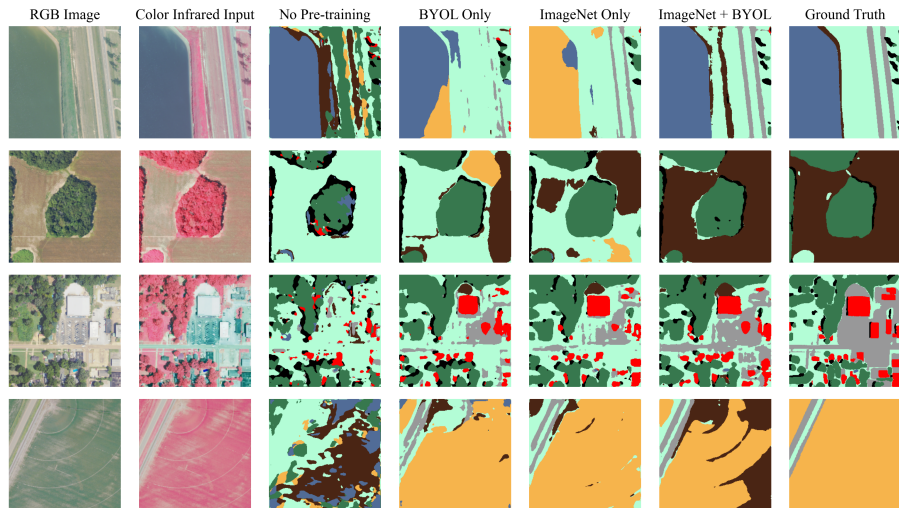


Figure 7: Using ImageNet and BYOL pre-training enables a model to produce classification maps of high visual fidelity under low training data scenarios.

Mississippi Land Cover Product

- Selected models trained using best combination of pre-training steps, architectures, and N_{train} from validation experiments to form ensemble for final inference (Marmanis et al., 2016).
 - ▶ ImageNet + BYOL pre-training
 - ▶ U-Net (Ronneberger et al., 2015)
 - ▶ $N_{\text{train}} = 750$
 - ▶ Ensemble of 4 models trained on different folds
- Sliding window inference with test-time augmentations (Moshkov et al., 2020) to generate land cover map for entire state of Mississippi from 2023 NAIP imagery.
- Use 25,000 randomly sampled, spatially independent points for validation and accuracy assessment

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Final Result: A 1 m Land Cover Map of Mississippi

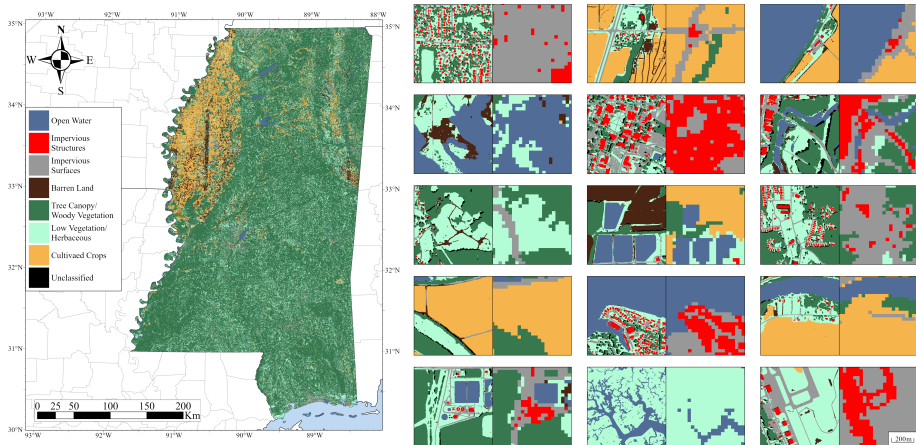


Figure 8: Our 1 m product provides far greater spatial fidelity compared to NLCD (30 m).

Confusion Matrix

Table 1: Confusion matrix for the Mississippi land cover classification results.

Predicted Label True Label	Open Water	Impervious Structures	Impervious Surfaces	Barren Land	Tree Canopy/ Woody Vegetation	Low Vegetation/ Herbaceous	Cultivated Crops	Unclassified	Producer's Accuracy
Open Water	511	1	0	10	11	18	0	21	89.34%
Impervious Structures	0	80	15	3	3	5	0	0	75.47%
Impervious Surfaces	1	5	183	17	1	22	0	1	79.57%
Barren Land	6	4	68	944	22	451	144	5	57.42%
Tree Canopy/ Woody Vegetation	1	2	2	6	14712	366	6	22	97.32%
Low Vegetation/ Herbaceous	7	2	20	138	376	3699	492	17	77.86%
Cultivated Crops	0	0	0	17	2	38	1243	1	95.54%
Unclassified	2	11	6	7	753	86	1	413	32.29%
User's Accuracy	96.78%	76.19%	62.24%	82.66%	92.64%	78.95%	65.91%	86.04%	Overall Accuracy 87.14%

Final mIoU = 62.77%

Conclusion

- High-quality, large-area VHSR land cover mapping is possible with very limited labeled data.
- **Key Takeaways:**
 - ① **Sample Smarter:** Semantic stratification using existing coarse data to guide sample selection can be used to build more informative training datasets.
 - ② **Pre-train Better:** Self-supervised learning on unlabeled domain-specific imagery is highly effective at bridging the domain gap between ImageNet and remote sensing imagery.
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Questions?

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